DATAOPS INNOVATION HACKATHON 2026

ARGUS

AI-Powered Unified VAPT Agent

Problem Statement 1

Conversational agent · real tools · human-approved

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github.com/RAHUL0408-B/DataOps

1. TEAM INTRODUCTION & PROBLEM STATEMENT

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DataOps Innovation Hackathon 2026 · Problem Statement 1: AI Agent for Automated Penetration Testing

Project: **ARGUS** - an AI-powered unified VAPT agent

THE PROBLEM STATEMENT, IN THE BRIEF'S OWN WORDS

"Design and build a conversational AI agent that guides an authorized security assessment from initial reconnaissance through the generation of a client-ready VAPT report... The agent must demonstrate actual security reasoning rather than blindly executing a predefined sequence of scanners."

2. THE PROBLEM

Penetration testing needs many tools - and even more manual work.

Recon, scanning, enumeration, vulnerability analysis, validation, reporting. Each stage needs a different tool, and analysts spend more time combining and interpreting outputs than actually analyzing security risk.

- Run nmap, Nikto, ZAP, Burp, Nessus, OpenVAS, Metasploit separately
- Manually correlate overlapping, conflicting results
- Hours of write-up before any real risk analysis begins

3. YOUR SOLUTION

"Scanners give you alerts. ARGUS gives you conclusions."

A conversational AI agent that plans, runs, and correlates a real penetration test - nmap, Nikto, OWASP ZAP and approval-gated Metasploit - then writes a client-ready report. Every finding carries evidence, a confidence level, and the reasoning behind it.

Evidence

Every finding traces back to saved raw tool output, hashed with SHA-256.

Confidence

A deterministic ledger scores every finding, with the reason written down.

Reasoning

A Decision Card for every step: what it saw, options weighed, why it chose.

A human approves every risky action. Always.

3. YOUR SOLUTION - HOW IT WORKS

Exactly the eight steps the brief asks for

01

Analyst Request

Plain-English target and scope

02

Agent Planning

Tool choice, with reasoning shown

03

Approved Orchestration

Real scans, only in scope

04

Result Ingestion

Raw tool output saved, hashed

05

AI Correlation

Duplicates merged, evidence-based

06

Risk Prioritization

A transparent, sourced formula

07

Human Review

Explicit approval for anything active

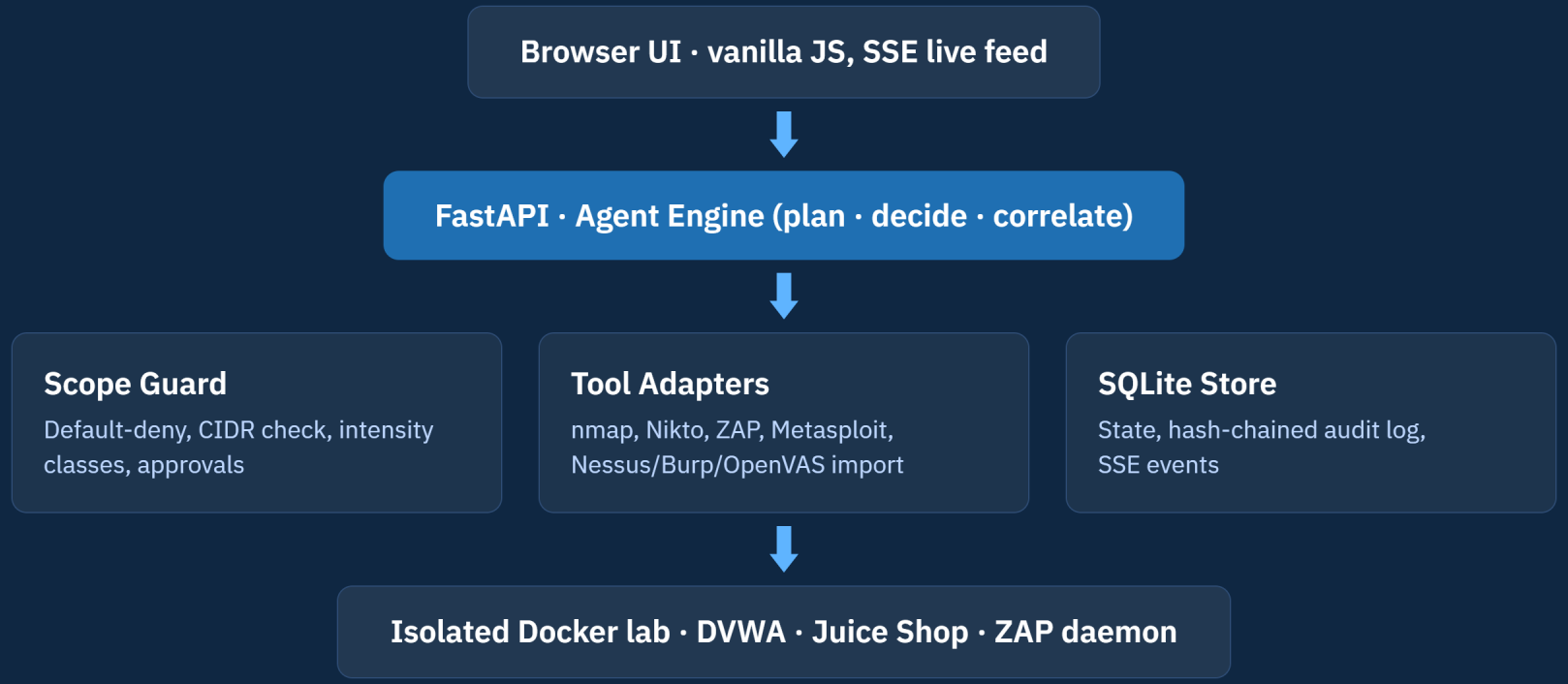
08

Unified VAPT Report

Client-ready PDF, every time

4. PROJECT ARCHITECTURE & STRUCTURE

Code enforces safety. The model only reasons.



5. TECH STACK

Software

Backend - Python, FastAPI, Pydantic v2, SQLite, httpx, reportlab

Frontend - Vanilla JavaScript, Server-Sent Events, hand-built SVG

Infrastructure - Docker Compose, isolated lab network

Optional AI - Google Gemini (free tier, verified live) or Anthropic Claude

Security tools integrated

Tool	Status
Nmap	Live
Nikto	Live
OWASP ZAP	Live
Metasploit	Live · approval-gated
Nessus	Import-only
Burp Suite	Import-only
OpenVAS / GVM	Import-only

Live = ARGUS runs the tool and parses real output. Import-only = commercial tools we don't have live licences for; ARGUS parses a report file instead of pretending to run them.

6. FEATURES

What ARGUS actually does, beyond running scanners

Adaptive AI reasoning

Every step writes a Decision Card: options weighed, evidence that triggered it, why it chose. An optional Gemini layer (verified live) adapts the investigation order per host.

Cross-tool correlation

3 independent tools reporting the same clickjacking issue become one finding, with all evidence kept and confidence scored, not 3 duplicates.

False-positive refutation

Config-type findings get one harmless re-check request; a genuinely wrong alert is auto-refuted with the reason recorded, not left for a human to eyeball.

Transparent risk scoring

priority = base × exposure × exploitability × criticality × confidence - every factor shown and sourced on the finding, never a bare number.

6. FEATURES (CONTINUED) - SAFETY & HUMAN-IN-THE-LOOP

The human stays in control. Always.

Default-deny scope

Out-of-scope targets are refused and logged - zero traffic sent, every time.

Explicit approval gate

Any active or intrusive action - ZAP active scan, Metasploit check - waits for a named analyst's approval.

Hash-chained audit

Every action logged; each entry's hash includes the one before it, so silent tampering is detectable.

Kill switch

One call stops every running job immediately; no new ones start afterward.

Metasploit: only 6 allow-listed, read-only auxiliary modules. No exploit modules, no payloads, never autonomous.

7. FEASIBILITY

Not a concept. It already runs, today, against a real target.

99/100
Live checks passed against the real Docker lab

34
Automated unit tests, all passing

SQLi + XSS
Genuinely found live, not staged

37
Page client-ready PDF report, generated from this run

Technically

Standard web stack, Docker for isolation, every security tool open-source except the commercial imports we're honest about. No exotic infrastructure.

Practically

Runs for $0 - open-source tools, free-tier AI. The AI layer is optional too: the deterministic planner works with zero API key, so adoption never depends on an LLM budget.

8. LIMITATIONS

Said plainly, not hidden.

⚠ **3 of 7 tools are import-only.** Nessus, Burp Suite and OpenVAS are commercial - we read their report files, we don't run them live, and we say so everywhere, not just here.

⚠ **The known-exploited-CVE list is a small, offline snapshot** - not a live feed. It boosts a handful of known-serious bugs; it won't catch everything.

⚠ **The Claude AI-planner path is unit-tested only** - no Anthropic API key was available to us. Gemini is the one verified against the live API.

⚠ **The default scope points at our own practice lab.** Pointing it at a different authorized target is one config change, not a code change - but it is a step someone has to take.

9. FUTURE SCOPE

Where we'd take this next.

Live OpenVAS, Nessus, Burp

With proper licences, upgrade the three import-only tools to live API integrations.

Richer confidence rules

Lower confidence when an independent tool re-tests and doesn't reproduce a finding; exposure factors beyond the current flat value.

Team workflows

Multi-analyst review, assignment, and comment threads on findings - not just a single approver.

Export formats

JSON/CSV export for SIEM and ticketing-system ingestion, alongside the PDF report.

Trend tracking

Compare assessments of the same target over time - what got fixed, what's new, what regressed.

Real-world deployment

Point it at an organizer-authorized target today - the architecture is ready now, not someday.